

Routing

Computer Networks

BSCS 6TH

LECTURE 08

Mr. Farooq Faisal

What is Router?

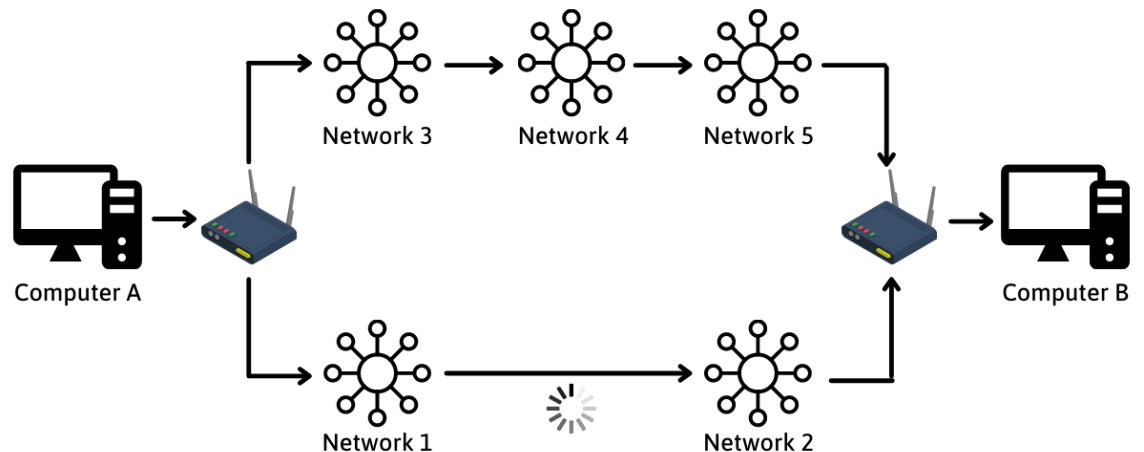
- ▶ A router is a specialized computer which can connect multiple networks to allow exchange of packets between them.
- ▶ Since a router uses IP header information (layer 3 protocol data unit), to transmit the packets between networks, this ability makes it a layer 3 device.
- ▶ Like switches, routers build a special database which serves as the source of information on what to do with incoming packets.
- ▶ This database is formally called Routing Information Base (RIB). But most often people call it a routing table.



What is Routing?

3

- ▶ Routing is the process of selecting paths in a network along which to send network traffic and route is the path to send the network traffic.
- ▶ There are two ways a router learn a route:
 - ▶ Static
 - ▶ Dynamic



Static Routing

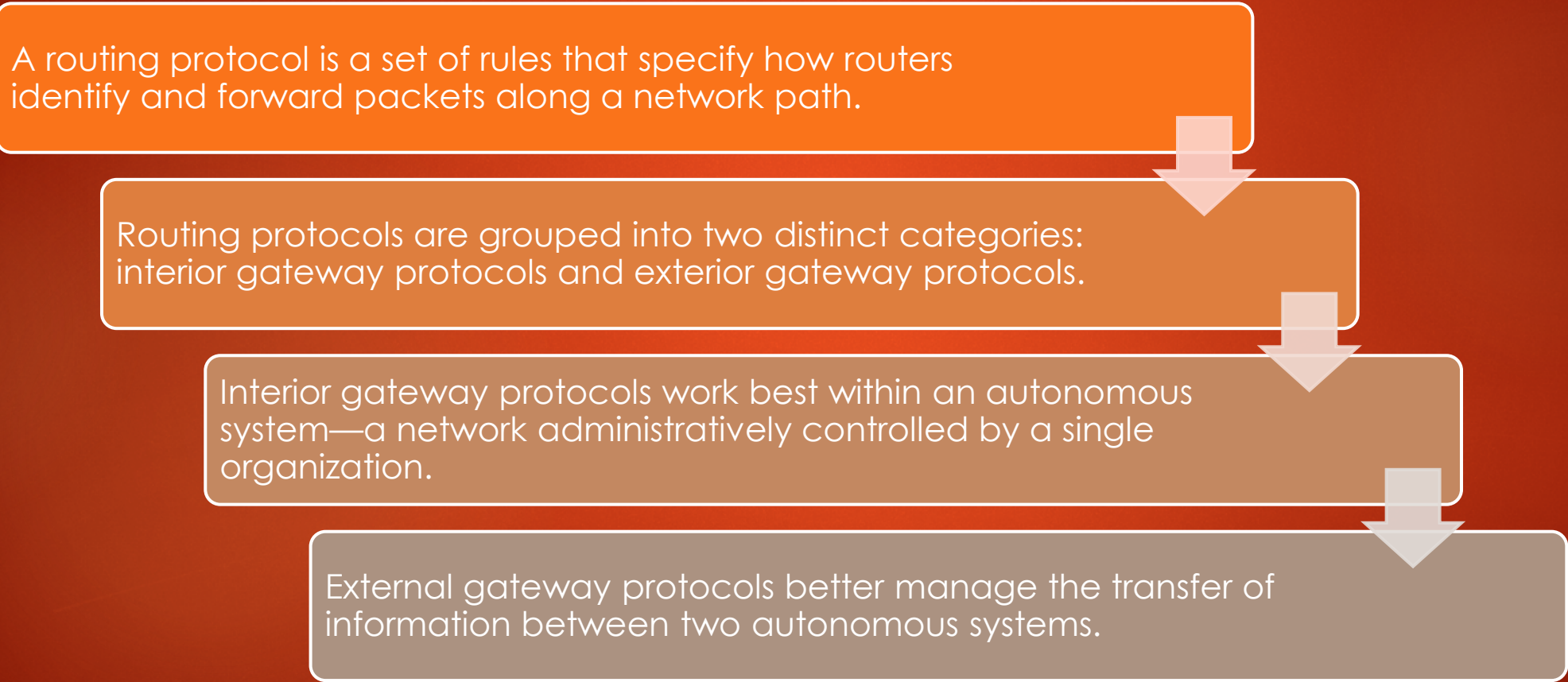
- ▶ A static route is a route that is manually configured on the router.
- ▶ Simply we can say a static route is a route that is created manually by a network administrator.
- ▶ Static routing uses small routing tables with only one entry for each destination.
- ▶ It also requires less computation time than dynamic routing because each route is preconfigured.
- ▶ Because static routes are preconfigured, administrators must manually reconfigure routes to adapt to changes in the network when they occur.
- ▶ Static routes are generally used in networks where administrators don't expect any changes.

Dynamic Routing

- ▶ Dynamic routing, sometimes called adaptive routing, is more complex than static routing because it creates more possible routes to send packets across a network.
- ▶ Dynamic routes are typically used in larger, fluid networks where static routes would be cumbersome to maintain and frequently reconfigure.
- ▶ Because dynamic routing is more complicated, it consumes more bandwidth than static routing.
- ▶ Dynamic routing uses algorithms to compute multiple possible routes and determine the best path for traffic to travel through the network.

Routing Protocols

A routing protocol is a set of rules that specify how routers identify and forward packets along a network path.



```
graph TD; A[A routing protocol is a set of rules that specify how routers identify and forward packets along a network path.] --> B[Routing protocols are grouped into two distinct categories: interior gateway protocols and exterior gateway protocols.]; B --> C[Interior gateway protocols work best within an autonomous system—a network administratively controlled by a single organization.]; C --> D[External gateway protocols better manage the transfer of information between two autonomous systems.];
```

Routing protocols are grouped into two distinct categories: interior gateway protocols and exterior gateway protocols.

Interior gateway protocols work best within an autonomous system—a network administratively controlled by a single organization.

External gateway protocols better manage the transfer of information between two autonomous systems.

Routing
Information
Protocol (RIP)

Interior Gateway
Protocol (IGRP)

Open Shortest
Path First (OSPF)

Exterior Gateway
Protocol (EGP)

Enhanced
Interior Gateway
Routing Protocol
(EIGRP)

Border Gateway
Protocol (BGP)

Intermediate
System-to-
Intermediate
System (IS-IS)

Static routing vs. dynamic routing

	Static routing	Dynamic routing
PATH SELECTION	One pre-configured route to destination	Multiple available routes to destination
ROUTE UPDATES	Engineers must reconfigure to make route changes	Algorithms automatically update with preferred route changes
ROUTING TABLES	Smaller routing table with only one entry for each destination	Routers send out entire routing tables to identify route availability
PROTOCOLS AND ALGORITHMS	Does not use protocols or algorithms for pre-configured route	Distance vector algorithms (RIP, IGRP) and link state algorithms (OSPF, IS-IS) adjust routes
COMPUTATION AND BANDWIDTH	Requires less computation time and bandwidth	Requires more computation and bandwidth
SECURITY	Better security	Less security
USE CASES	Used in smaller networks with fewer routers and unchanging network architecture	Used in larger networks and in networks that change frequently



**Any
Question?**

Thank you